

# Generator-to-Residence Barrier Noise Analysis

Outdoor sound propagation, path-length difference, and barrier sensitivity study

## Core question

Using a publicly documented octave-band sound-power spectrum for a commercial generator, what sound-pressure level is predicted at a residential property line, and how would an intervening barrier change the octave-band and overall dBA results?

## Current status

Project plan established August 1, 2026. Source selection, workbook implementation, calculations, and conclusions remain pending.

## Purpose

Create a specific, auditable portfolio exercise resembling an entry-level environmental-noise consulting task. The project replaces a broader source-combination calculator concept with one defined source, receiver, mitigation element, and decision-oriented parametric study.

## Professional origin

Acoustical consultant Ray Plasse proposed the generator-to-residence barrier scenario and emphasized path-length difference, receiver-level prediction, barrier geometry, and optional environmental attenuation effects. The guidance defines the direction; it does not imply Ray's supervision, review, endorsement, or responsibility for the work.

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# 1. Case Definition and Scope

Version 1 will model one stationary commercial generator and one residential property-line receiver. Published manufacturer or technical-document octave-band sound-power data will replace field measurements. The study will predict a baseline receiver level, introduce one solid barrier, and compare how barrier height and position change predicted attenuation.

## Required inputs

- Generator make/model, operating condition, rated load, and traceable public source document.
- Octave-band sound-power levels ( $L_w$ ) with units and reference conditions.
- Source height, receiver height, horizontal source-receiver distance, and property-line definition.
- Barrier height, source-to-barrier distance, barrier-to-receiver distance, and assumption of sufficient width/no gaps.
- A-weighting corrections and one documented outdoor-propagation/barrier method.

## Version 1 boundaries

- Include geometric spreading and barrier attenuation.
- Defer ground absorption, air absorption, atmospheric effects, directivity, reflections, terrain, multiple barriers, background correction, and regulatory compliance until the core model is stable.
- Label every output as an educational prediction, not a measurement or professional compliance assessment.
- Use one consistent method and unit system; do not combine equations from incompatible sources without justification.

## Conceptual calculation chain

| Stage                 | Output                                               |
|-----------------------|------------------------------------------------------|
| Published source data | Octave-band $L_w$ spectrum                           |
| Baseline propagation  | Unmitigated octave-band $L_p$ at receiver            |
| Barrier geometry      | Direct path, diffracted path, path-length difference |
| Barrier method        | Frequency-dependent insertion loss                   |
| A-weighting + dB sum  | Baseline and mitigated overall dBA                   |
| Parameter sweep       | Reduction versus barrier height and position         |

## 2. Workbook Architecture

| Worksheet                       | Purpose                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Read Me and Assumptions</b>  | Purpose, definitions, units, editable inputs, method references, limitations, and version status.                  |
| <b>Published Generator Data</b> | Model, operating condition, octave-band Lw, URL, page/table, date accessed, and transcription check.               |
| <b>Geometry and Scenarios</b>   | Source/receiver heights and locations, barrier dimensions and position, named baseline and mitigation cases.       |
| <b>Propagation</b>              | Distance, geometric divergence, baseline octave-band Lp, and later optional attenuation terms.                     |
| <b>Barrier Calculation</b>      | Direct and diffracted paths, path-length difference, wavelength/Fresnel inputs, insertion loss, and method limits. |
| <b>A-Weighted Results</b>       | Unmitigated and mitigated octave-band Lp, corrections, overall dBA, and reduction.                                 |
| <b>Sensitivity Dashboard</b>    | Barrier-height and placement comparisons with tables and graphs.                                                   |
| <b>Validation</b>               | Hand checks, limiting cases, unit checks, independent comparison, and formula-error status.                        |

### Method-control rule

The workbook will display equations, units, references, and intermediate values. The barrier equation will be implemented only after the Miller noise-control material recommended by Ray, or another authoritative method, is obtained and its assumptions and validity range are recorded.

### 3. Development Plan

#### Phase 0 - Plan and source strategy

Lock the case definition and staged boundaries; identify candidate CAT, Cummins, or comparable generator datasets; locate the relevant outdoor-propagation/barrier reference; select and document one calculation convention.

#### Phase 1 - Baseline propagation

Transcribe and verify one generator Lw spectrum; define representative property-line geometry; calculate baseline octave-band Lp and overall dBA without a barrier.

#### Phase 2 - Barrier effect

Calculate direct and diffracted path lengths and path-length difference; apply the selected barrier method by band; compare baseline and mitigated levels.

#### Phase 3 - Parametric study

Sweep barrier height; compare positions nearer the source, midpoint, and receiver while keeping total source-receiver distance fixed; identify diminishing returns and frequency dependence.

#### Phase 4 - Optional refinements

Add ground, air, or atmospheric attenuation one term at a time, documenting the source and validity range and comparing each refinement with the core model.

#### Phase 5 - Reviewable evidence

Complete validation and limitations; export a clean dashboard and geometry figure; write a short consulting-style technical memorandum.

#### August 1 objective

**Finalize this plan only: define the case, phases, required inputs, source-data strategy, deliverables, validation checks, and scope boundaries. No workbook, numerical result, or consulting conclusion is required today.**

## 4. Validation, Deliverables, and Review

### Validation requirements

- Confirm copied source spectra against the cited document and independently recompute any published overall value when possible.
- Check feet-to-meters conversion, frequency-band labels, and the distinction between Lw and Lp.
- Hand-calculate at least one baseline band and one barrier case.
- Test no-barrier and zero/near-zero path-difference behavior against the chosen method's stated limits.
- Confirm logarithmic summation and A-weighting with a second calculation method.
- Record discrepancies and unresolved assumptions instead of adjusting results to look plausible.

### Planned portfolio evidence

- Reusable Excel workbook with visible formulas and citations.
- Source-barrier-receiver geometry diagram.
- Baseline and mitigated octave-band tables and overall dBA comparison.
- Barrier reduction versus height graph; placement graph if the core model is stable.
- Short consulting-style memorandum with assumptions, method, results, recommendations, and limitations.

### Questions for a knowledgeable reviewer

- Is the initial scope appropriate for demonstrating entry-level consulting reasoning?
- Is the proposed propagation/barrier method appropriate for this simplified geometry?
- Which assumptions or checks are most important before treating the results as meaningful?
- Would barrier height, barrier placement, or another variable make the strongest sensitivity study?
- What would make the final memorandum resemble useful junior-consultant work without overstating competence?

### Scope statement

This plan establishes a learning and portfolio exercise. It does not establish that the model has been implemented, validated, reviewed, or used for a real client or regulatory decision.